

High-Frequency Trading in a Limit Order Book

Optimization Methods in Finance — FS26

Harjot Singh

May 19, 2026

- **Problem Statement:** Dealers must continuously post bid and ask quotes to provide liquidity while managing the risk of holding a volatile asset.
- **Inventory Risk:** Arises from uncertainty in the asset's terminal value, directly impacting the dealer's terminal wealth.
- **Approach:** Combine a utility-based indifference pricing framework with empirically grounded order execution intensities derived from econophysics.

- **Agent State:** Defined by current wealth in cash, inventory in shares, and the asset's mid-price.
- **Objective:** Maximize expected exponential utility of terminal wealth at a finite horizon.
- **Controls:** The agent controls the continuous placement of bid and ask limit orders at specified distances from the mid-price.
- **Market Interaction:** Limit orders are executed by incoming market orders according to Poisson processes with intensities dependent on quote distances.

x	Initial wealth in dollars.
q	Inventory of stocks held.
T	Terminal time.

- **Process:** The mid-market price evolves as an arithmetic Brownian motion with zero drift.
- **Assumption:** The agent has no opinion on the drift or autocorrelation structure of the stock.
- **Utility Boundedness:** Arithmetic Brownian motion is chosen over Geometric Brownian motion to ensure exponential utility functionals remain bounded.
- **SDE:**

$$dS_u = \sigma dW_u$$

- **Moments:** Under the natural filtration $\mathcal{F}_t$, for $T > t$:

$$E_t[S_T] = S_t = s$$

$$\text{Var}_t(S_T) = \sigma^2(T - t)$$

S_t, s	Mid-market price of the stock at time t .
σ	Constant volatility of the mid-price.
W_t	Standard one-dimensional Brownian motion.

- **Utility Function:** The agent uses a convex exponential utility measure, which implies constant absolute risk aversion (CARA).
- **Objective Formulation:** Maximize expected utility of the final Profit and Loss (P&L) profile at time T .
- **Terminal Wealth:** Consists of cash X_T plus the liquidation value of inventory $q_T S_T$.
- **Value Function:**

$$u(s, x, q, t) = \max_{\delta^a, \delta^b} E_t[-\exp(-\gamma(X_T + q_T S_T))]$$

γ	Risk aversion parameter.
X_T, x	Cash wealth at terminal time T , and initial cash.
q_T	Inventory at time T .
δ^a, δ^b	Distances from mid-price for ask and bid quotes.

- **Scenario:** An inactive trader holding inventory q until T without quoting limit orders.
- **Terminal Random Variable:** $S_T \sim N(s, \sigma^2(T - t))$.
- **Derivation:** Using the Gaussian MGF $E[\exp(cY)] = \exp(c\mu + \frac{1}{2}c^2\Sigma^2)$:

$$\begin{aligned}v(x, s, q, t) &= E_t[-\exp(-\gamma(x + qS_T))]
 &= -\exp(-\gamma x)E_t[\exp(-\gamma qS_T)]
 &= -\exp(-\gamma x)\exp\left(-\gamma qs + \frac{1}{2}(\gamma q)^2\sigma^2(T - t)\right)\end{aligned}$$

- **Final Expression:**

$$v(x, s, q, t) = -\exp(-\gamma x)\exp(-\gamma qs)\exp\left(\frac{\gamma^2 q^2 \sigma^2 (T - t)}{2}\right)$$

$v(x, s, q, t)$ Value function for the frozen inventory scenario.

- **Definition:** The price that makes the agent indifferent between their current portfolio and the portfolio ± 1 stock.
- **Bid Indifference:** $v(x - r^b(s, q, t), s, q + 1, t) = v(x, s, q, t)$
- **Derivation for r^b :** Equate the exponents of the frozen value function:

$$-\gamma(x - r^b) - \gamma(q + 1)s + \frac{1}{2}\gamma^2(q + 1)^2\sigma^2(T - t) = -\gamma x - \gamma qs + \frac{1}{2}\gamma^2 q^2 \sigma^2(T - t)$$

$$\gamma r^b - \gamma s + \frac{1}{2}\gamma^2\sigma^2(T - t)(q^2 + 2q + 1 - q^2) = 0$$

$$r^b(s, q, t) = s + (-1 - 2q)\frac{\gamma\sigma^2(T - t)}{2}$$

- **Indifference Price (Average):**

$$r(s, q, t) = \frac{r^a + r^b}{2} = s - q\gamma\sigma^2(T - t)$$

r^b, r^a Reservation bid and ask prices.

$r(s, q, t)$ Mid-point reservation or indifference price.

- **Mechanism:** The agent quotes limit orders at prices p^b and p^a .
- **Distances:** Formulated as distances from the mid-price:

$$\delta^b = s - p^b \quad \text{and} \quad \delta^a = p^a - s$$

- **Execution Processes:** Executions occur via Poisson point processes N_t^a (sell limit orders lifted) and N_t^b (buy limit orders hit).
- **Wealth Dynamics:** Cash wealth jumps upon execution:

$$dX_t = p^a dN_t^a - p^b dN_t^b$$

- **Inventory Dynamics:** Inventory jumps correspondingly:

$$q_t = N_t^b - N_t^a$$

p^b, p^a	Bid and ask limit order prices quoted by the agent.
N_t^b, N_t^a	Poisson processes for shares bought and sold.

- **Market Impact Theory:** Intensity models the probability a limit order is hit by a market order.
- **Inputs:** Order size density $f^Q(x) \propto x^{-1-\alpha}$ and log impact $\Delta p \propto \ln(Q)$.
- **Deriving Exponential Intensity:**

$$\begin{aligned}\lambda(\delta) &= \Lambda P(\Delta p > \delta) \\ &= \Lambda P(\ln(Q) > K\delta) = \Lambda P(Q > \exp(K\delta)) \\ &= \Lambda \int_{\exp(K\delta)}^{\infty} x^{-1-\alpha} dx = \frac{\Lambda}{\alpha} \exp(-\alpha K\delta)\end{aligned}$$

$$\lambda(\delta) = A \exp(-k\delta) \quad (\text{where } A = \Lambda/\alpha \text{ and } k = \alpha K)$$

$\lambda(\delta)$	Poisson intensity of execution vs. distance.
Λ, α	Constant frequency of orders, tail exponent of size distribution.
A, k	Constants for the exponential intensity function.

- **Dynamic Programming:** Optimal value function u satisfies the HJB equation.
- **HJB Equation:**

$$u_t + \frac{1}{2}\sigma^2 u_{ss} + \max_{\delta^b} \lambda^b(\delta^b)[u(s, x - s + \delta^b, q + 1, t) - u(s, x, q, t)] \\ + \max_{\delta^a} \lambda^a(\delta^a)[u(s, x + s + \delta^a, q - 1, t) - u(s, x, q, t)] = 0$$

- **Terminal Condition:** $u(s, x, q, T) = -\exp(-\gamma(x + qs))$.
- **Ansatz Substitution:** Assume $u = -\exp(-\gamma x) \exp(-\gamma \theta(s, q, t))$.
- **Derivation (Canceling x):** Substitute into the jump term for δ^b :

$$-\exp(-\gamma(x - s + \delta^b - \theta(s, q + 1, t))) - (-\exp(-\gamma(x + \theta(s, q, t))))$$

Divide entirely by $-\gamma \exp(-\gamma x) \exp(-\gamma \theta(s, q, t))$ to obtain the reduced PDE for θ .

u_{ss}, u_t	Partial derivatives of the value function.
$\theta(s, q, t)$	Reduced state value function independent of cash wealth.

- **Reduced PDE for θ :** Contains the optimized jump terms.

$$\max_{\delta^b} \left[\frac{\lambda^b(\delta^b)}{\gamma} (1 - e^{\gamma(s - \delta^b - r^b)}) \right]$$

- **First-Order Condition:** Differentiate with respect to δ^b :

$$\frac{\partial \lambda^b}{\partial \delta^b} \frac{1}{\gamma} (1 - e^{\gamma(s - \delta^b - r^b)}) + \frac{\lambda^b}{\gamma} (\gamma e^{\gamma(s - \delta^b - r^b)}) = 0$$

- **Implicit Equation Rearrangement:**

$$1 - e^{\gamma(s - \delta^b - r^b)} = -\gamma \frac{\lambda^b}{\partial \lambda^b / \partial \delta^b} e^{\gamma(s - \delta^b - r^b)}$$

$$e^{-\gamma(s - \delta^b - r^b)} - 1 = -\gamma \frac{\lambda^b}{\partial \lambda^b / \partial \delta^b}$$

$$s - r^b(s, q, t) = \delta^b - \frac{1}{\gamma} \ln \left(1 - \gamma \frac{\lambda^b(\delta^b)}{\partial \lambda^b / \partial \delta^b} \right)$$

$$\frac{\partial \lambda^b}{\partial \delta^b}$$

Sensitivity of the execution intensity to the bid distance.

- **Intuitive Pricing Logic:** The complex PDE implies a two-step algorithmic procedure for the market maker.
- **Step 1: Compute Indifference Price:** Solve the PDE for θ to extract the reservation bid and ask prices $r^b(s, q, t)$ and $r^a(s, q, t)$ based purely on inventory risk.
- **Step 2: Calibrate to Market:** Solve the implicit equations for δ^a, δ^b to calibrate quotes to the limit order book's supply λ^b and demand λ^a .
- **Synthesis:** The strategy fundamentally balances personal risk considerations (the indifference price shifting with inventory) against the market environment (the execution probability curve).

- **Challenge:** The PDE for θ has highly nonlinear arrival terms depending on q .
- **Expansion:** Asymptotic expansion in the inventory variable:

$$\theta(q, s, t) = \theta^0(s, t) + q\theta^1(s, t) + \frac{1}{2}q^2\theta^2(s, t) + \dots$$

- **Linear Approximation:** With $\lambda(\delta) = Ae^{-k\delta}$, approximate the arrival terms:

$$\frac{A}{k + \gamma}(e^{-k\delta^a} + e^{-k\delta^b}) \approx \frac{A}{k + \gamma}(2 - k(\delta^a + \delta^b))$$

- **Grouping terms:** $\mathcal{O}(q)$ yields $\theta^1(s, t) = s$, and $\mathcal{O}(q^2)$ yields $\theta^2(s, t) = -\frac{1}{2}\sigma^2\gamma(T - t)$.
- **Approximate Spread Formula:**

$$\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right)$$

$\theta^1(s, t)$ Reservation price when inventory is zero.

$\theta^2(s, t)$ Sensitivity of market maker's quotes to inventory changes.

- **Infinite Horizon Utility:** To remove time-dependence ($T - t$), consider an infinite horizon objective measuring stationary reservation prices.

$$\bar{v}(x, s, q) = E \left[\int_0^\infty -\exp(-\omega t) \exp(-\gamma(x + qS_t)) dt \right]$$

- **Stationary Reservation Ask:**

$$\bar{r}^a(s, q) = s + \frac{1}{\gamma} \ln \left(1 + \frac{(1 - 2q)\gamma^2 \sigma^2}{2\omega - \gamma^2 q^2 \sigma^2} \right)$$

- **Parameter Constraint:** Requires $\omega > \frac{1}{2}\gamma^2 \sigma^2 q^2$ to keep utility bounded.
- **Interpretation:** ω acts as an upper bound on allowable inventory position q_{max} .

ω	Discount factor / upper bound parameter for stationary inventory.
$\bar{r}^a(s, q)$	Stationary reservation ask price.

- **Time Step:** Choose dt large enough to exceed typical tick time but small enough to limit multiple orders per step.
- **Pseudocode:**
 - ① At time t , compute r , δ^a , and δ^b based on state $s_t, q_t, T - t$.
 - ② Compute execution probabilities: $P_a = \lambda^a(\delta^a)dt$ and $P_b = \lambda^b(\delta^b)dt$.
 - ③ Draw uniformly distributed random variables $U_a, U_b \sim U[0, 1]$.
 - ④ If $U_a < P_a$: $q_{t+dt} = q_t - 1$, $x_{t+dt} = x_t + s_t + \delta^a$.
 - ⑤ If $U_b < P_b$: $q_{t+dt} = q_t + 1$, $x_{t+dt} = x_t - s_t + \delta^b$.
 - ⑥ Update mid-price: $s_{t+dt} = s_t \pm \sigma\sqrt{dt}$.
- **Complexity:** Linear time complexity $\mathcal{O}(N)$ where $N = T/dt$ steps.

dt	Simulation time step.
U_a, U_b	Standard uniform random variables.

- γ (**Risk aversion**): Higher γ widens the spread and increases the quote skew per unit of inventory.
- σ (**Volatility**): Higher σ increases inventory risk, directly widening the spread via $\gamma\sigma^2(T - t)$.
- k (**Order flow elasticity**): Higher k (steeper intensity drop-off) tightens the market microstructure component of the spread.
- $T - t$ (**Time to horizon**): Spread narrows as $t \rightarrow T$ because residual inventory risk vanishes.

- **Benchmark:** The "Symmetric strategy" uses the optimal average spread but strictly centers it around the mid-price regardless of inventory.
- **Performance ($\gamma = 0.1$):**
 - *Symmetric Strategy:* Receives higher volume, slightly higher average profit (\$68.4).
 - *Inventory Strategy:* Lower average profit (\$65.0) but significantly lower standard deviation in profit (\$6.6 vs \$12.7) and final inventory.
- **Risk Aversion Impact:**
 - As $\gamma \rightarrow 0$: The inventory effect vanishes, and the strategies become identical.
 - High γ ($\gamma = 1$): The inventory strategy aggressively quotes to avoid accumulating positions, strictly suppressing variance at the cost of expected profit.

- **Zero Drift Assumption:** The Brownian motion has no drift. The model ignores short-term alpha signals or momentum, which are vital in real high-frequency environments.
- **Exponential Intensity:** The reliance on pure exponential decay $\lambda(\delta) = Ae^{-k\delta}$ ignores real-world limit order book queueing dynamics and partial fills.
- **No Latency:** The model assumes continuous, costless updates of p^b, p^a with zero latency.
- **Single Share Execution:** Assumes deterministic jump sizes of exactly 1 share per execution, ignoring multi-share lot impacts.

- **Queue Position:** Modify state space to include queue priority, changing $\lambda(\delta)$ to depend on outstanding volume ahead of the agent.
- **Hawkes Processes:** Replace Poisson arrivals N_t with Hawkes processes to capture the empirically observed clustering of market orders.
- **Alpha/Drift:** Inject a short-term predictive drift μ_t into $dS_t = \mu_t dt + \sigma dW_t$. The HJB must then account for directional skewing.
- **Latency Delay:** Introduce a control delay τ . Quotes δ_t^a apply to state variables at $t - \tau$, moving the framework into stochastic delay differential equations.

- **Core Model:** Dealer maximizes $E_t[-\exp(-\gamma(X_T + q_T S_T))]$ via jump-diffusion controls.

- **Indifference Price:**

$$r(s, q, t) = s - q\gamma\sigma^2(T - t)$$

Adjust valuation downward if long inventory, upward if short.

- **Optimal Spread:**

$$\delta^a + \delta^b \approx \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln\left(1 + \frac{\gamma}{k}\right)$$

- **Practical Rules:**

- 1 Base pricing on subjective inventory risk, not just mid-price.
- 2 Widen spread proportionally to volatility and your risk aversion.
- 3 Shift quotes aggressively to flatten inventory as holding risk increases.

• Mean/Variance Objective Derivation:

- Alternative objective: $V(x, s, q, t) = E_t[(x + qS_T) - \frac{\gamma}{2}(qS_T - qs)^2]$.
- Under geometric Brownian motion: $\frac{dS_u}{S_u} = \sigma dW_u$.
- $E[S_T^2] = s^2 \exp(\sigma^2(T - t))$.
- $V(x, s, q, t) = x + qs - \frac{\gamma q^2 s^2}{2}(e^{\sigma^2(T-t)} - 1)$.
- Yields reservation ask: $R^a(s, q, t) = s + \frac{(1-2q)}{2}\gamma s^2(e^{\sigma^2(T-t)} - 1)$.

• Power Law Intensity Algebraic Note:

- If $\Delta p \propto Q^\beta$, substituting into $P(Q > x)$ gives $\lambda(\delta) = B\delta^{-\alpha/\beta}$. HJB substitution requires numerical routines.

$V(x, s, q, t)$	Mean/variance objective function.
$R^a(s, q, t)$	Reservation ask under GBM mean/variance.

- Avellaneda, M. and Stoikov, S., High-frequency trading in a limit order book. *Quantitative Finance*, 2008, 8, 217-224.
- Ho, T. and Stoll, H., Optimal dealer pricing under transactions and return uncertainty. *J. Financ. Econ.*, 1981, 9, 47-73.
- Gabaix, X., Gopikrishnan, P., Plerou, V. and Stanley, H.E., Institutional investors and stock market volatility. *Quart. J. Econ.*, 2006, 121, 461-504.